CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (CL) Examination

May 2012

CL 207 Concrete Technology

Date: 14 .05.2012, Monday | Time: 01:30 a.m. To 04:30 p.m. | Maximum Marks: 70

Instructions:
1. The question paper comprises of two sections. Read whole question paper then attempt.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1	(a)	State the necessary qualities of fresh water used for making good concrete.	01
	(b)	Define workability and describe factors affecting the same.	03
	(c)	What is the effect of age on strength of concrete?	03
Q - 2	(a)	Explain classification of aggregate? State the importance of each in making of concrete.	03
	(b)	What is alkali aggregate reaction? State the factors affecting the same.	04
	(c)	Describe various steps involved in manufacturing process of fresh concrete. Briefly explain each.	07
		OR	
Q - 2	(a)	State the factors affecting choice of aggregate in concrete. Explain any one of them.	03
	(b)	Explain laboratory test to find out aggregate crushing value. Give your comments if aggregate crushing value is 40.	04
	(c)	Why curing is important? Compare the hessian cloth and pounding method of curing. Discuss their result with respect to compressive strength at various W/C ratios.	04
	(d)	Enlist the methods of measurement of workability and describe anyone in brief.	03
Q - 3	(a)	What are admixtures? Enlist admixtures used in concrete. State the benefits and limitations of using the admixtures in concrete.	07
	(b)	What is gel-space ratio? Mention the graphical co-relation of compressive strength of mortar and gel-space ratio.	03
	(c)	Write short note on rebound hammer test including use, procedure of test, result interpretation from reading and limitation of the test.	04
		OR	
Q - 3	(a)	Explain Plasticizer in brief and its role in concrete.	03
	(b)	State effects and use of super plasticizer in concrete.	04
	(c)	Explain the causes of corrosion of steel in RCC. Explain carbonation.	07

SECTION - II

Q - 4 (a) Proportion the concrete mix for RCC with following design stipulation. Provide result for 1 bag of cement as per IS 10262-2009. 07

Minimum compressive strength of concrete at 28 day		30 MPa
Maximum size of aggregate		20 mm, Angular
Slump value		50 mm
Cement		OPC – 53 grade
Degree of control		Good
Exposure condition		Severe
Test data		Sp. Gravity
Cement		3.15
FA- Zone II	all in one grading	2.6
CA-Sieve analysis		2.7
Water		1

Q - 5 (a) Name various method of proportioning of concrete. 02

(b) Enlist the ingredients of modern concrete. Explain the role of each ingredient. 05

(c) What is heat of hydration? Explain the role of water requirement in hydration process. 03

(d) What is consistency of cement? Explain standard consistency test on cement and its significance in measuring the other property. 04

OR

Q – 5 (a) Maximum size of aggregate, grading, surface texture and shape of coarse aggregate govern the water demand of concrete. Justify the statement. 02

(b) Classify the concrete as per IS: 456-2000. State disadvantage of concrete. 05

(c) Explain the process of hydration of cement. 03

(d) Explain the dry manufacturing process of Portland cement with neat sketch. 04

Q – 6 (a) Explain how will you account for the moisture present in sand while mix proportioning? 02

(b) Write short note on polymer concrete. 05

(c) State the properties and use of High Alumina Cement. 03

(d) Explain the production of fly ash. State its effect on fresh concrete. 04

OR

Q – 6 (a) What are the acceptance criteria of concrete as per IS: 456-2000? 02

(b) Write a short note on fiber reinforced concrete. 05

(c) What are the pozzolonic admixtures? Enlist them and explain any one in detail. 03

(d) State the properties and use of sulphate resisting cement. 04

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